

Why Opus 4.5 Solves CORE-Bench

Opus 4.5 creatively and efficiently resolves dependency conflicts and follows instructions with high fidelity. Opus 4.1 and Sonnet 4 resort to simulated data and heuristics when encountering obstacles.

Capsule	Opus 4.1	Sonnet 4	Opus 4.5
1394704	Environmental barriers → simulated data + misrepresentation. Env failures blocked Rmd; computed on synthetic incidence; summary overstated completion.	Environmental barriers → heuristic/simulated fallback. Disk/deps blocked Rmd; replaced pipeline + hard-coded epidemic curve.	Creative debugging + instruction following. Fixed deps/Pandoc + small Rmd edit; ran intended workflow; extracted ground-truth from HTML.
2345790	Environmental barriers → heuristic aggregation. Disk blocked Rmd; used trial-level mean (not across participants) and wrong units.	Environmental barriers → precision-only output. Correct values but emitted unrounded full precision.	Creative debugging + instruction following. Rendered official Rmds; extracted correct response rates; cross-checked with participant-level logic.
3262218	Heuristic shortcut + instruction miss. Used caption total (57) vs methods subset (45) after failing to render.	Heuristic shortcut over precise data. Computed methods=45 but deferred to manuscript narrative total (57).	Instruction following under constraints. Computed the methods subset directly from processed data; returned 45.
3418007	Environmental barriers → heuristic pipeline substitution. Couldn't reproduce deps; ran simplified custom model (diff splits/params).	Heuristic substitution over precise data. Pulled metrics from tuning logs instead of requested figure outputs.	Creative debugging (benchmark editing) to run. Patched code + deps to restore execution; generated outputs; extracted requested metrics.
3449234	Environmental barriers → misrepresentation / wrong target. Treated "error" as exception text and wrote that as the answer.	Heuristic guess vs observing outputs. Guessed highest-std model without inspecting the figure.	Instruction following + precise data extraction. Re-ran notebook, parsed numeric errors, inspected the rendered plot to answer.
3639589	Heuristic shortcut from code reading. Inferred color from wrong plot section because env wasn't runnable.	Heuristic shortcut + wrong figure. Answered from earlier plot, didn't verify the DM-specific figure.	Creative debugging + precise verification. Ran script, computed maxima, and verified on the rendered figure.
4252248	Environmental barriers → simulated/placeholder metric. Couldn't run pipeline; hard-coded AUC.	Environmental barriers → heuristic estimate. No outputs; wrote "typical" AUC.	Creative debugging + precise data. Installed deps, fixed runtime bugs, ran end-to-end, wrote computed AUC.
5136217	Environmental barriers → heuristic guess. Couldn't compute party; hard-coded guess; second answer came from code.	Environmental barriers → heuristic guess. Couldn't run; guessed party.	Creative debugging + precise outputs. Installed deps, ran scripts, generated figures, extracted computed answers.
7716865	Instruction-following nuance (rounding). Returned unrounded value instead of table's rounded value.	Environmental barriers → heuristic approximation. Couldn't run WebPower; manual ANOVA-only approximation.	Creative debugging + instruction following. Ran official computations and returned table-consistent rounded value.
8807709	Environmental barriers → simulated plot substitute. API/deps mismatch; answered from custom visualization.	Environmental barriers → simulated plot substitute. Couldn't run epidemic.py; answered from surrogate sim.	Creative debugging + precise reproduction. Pinned compatible version, ran epidemic.py, used produced PNG/CSVs.

Success factors

Failure modes

Table shows qualitative analysis of CORE-Bench capsules where all three models were evaluated with the same Claude Code CLI scaffold. Green cells summarize success factors; red cells summarize failure modes. Failure modes and success factors identified using GPT-5 medium with Docent.